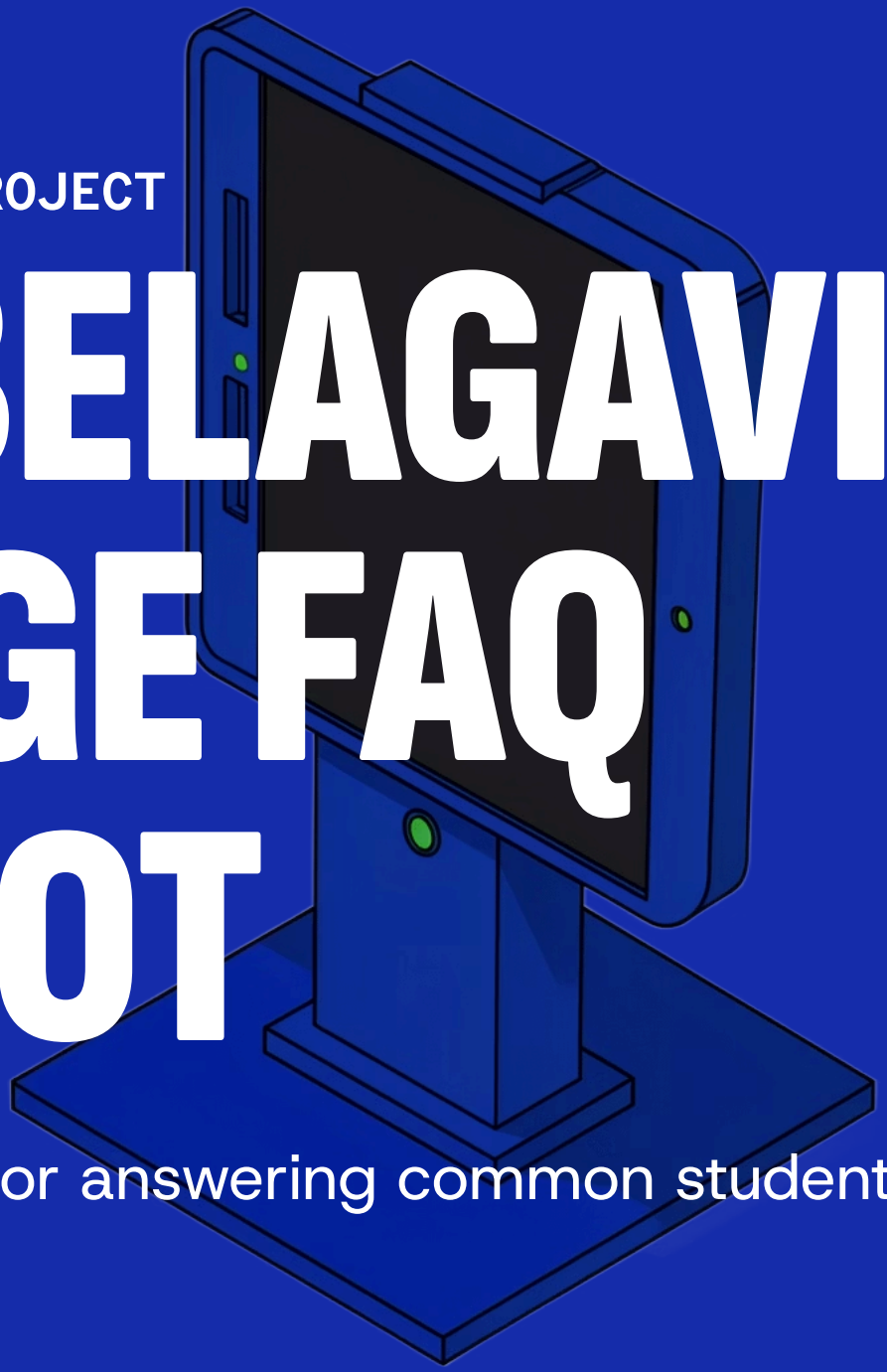
GTTC BELAGAVI · NLP PROJECT

GTTC BELAGAVI COLLEGE FAQ CHATBOT



A practical NLP project for answering common student questions quickly.

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WHY IT MATTERS

STUDENT FRIENDLY

1 **College enquiries repeat.** Courses, admission, fees, timings, examinations, placements, and contact details.

2 **A simpler conversation.** The project turns those recurring questions into a simple conversational experience.

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SYSTEM IDEA

THE CHATBOT COMBINES RULES WITH A FOCUSED FAQ KNOWLEDGE BASE

01

Ask

Students ask questions in a web interface.

02

Match

The system matches keywords and patterns to a likely intent.

03

Answer

A predefined answer is returned, with a fallback when the question is not recognized.

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NLP PIPELINE

ONE QUESTION PASSES THROUGH EIGHT NLP STEPS

01 User input

02 Tokenization

03 Stopword handling

04 Pattern matching

05 Intent detection

06 FAQ search

07 Response generation

08 NLP analysis display

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WEB INTERFACE

THE INTERFACE MAKES COLLEGE INFORMATION EASY TO EXPLORE

Category shortcuts organize timings, courses, AIML, admission, fees, training, contact, location, and placements.

Timings

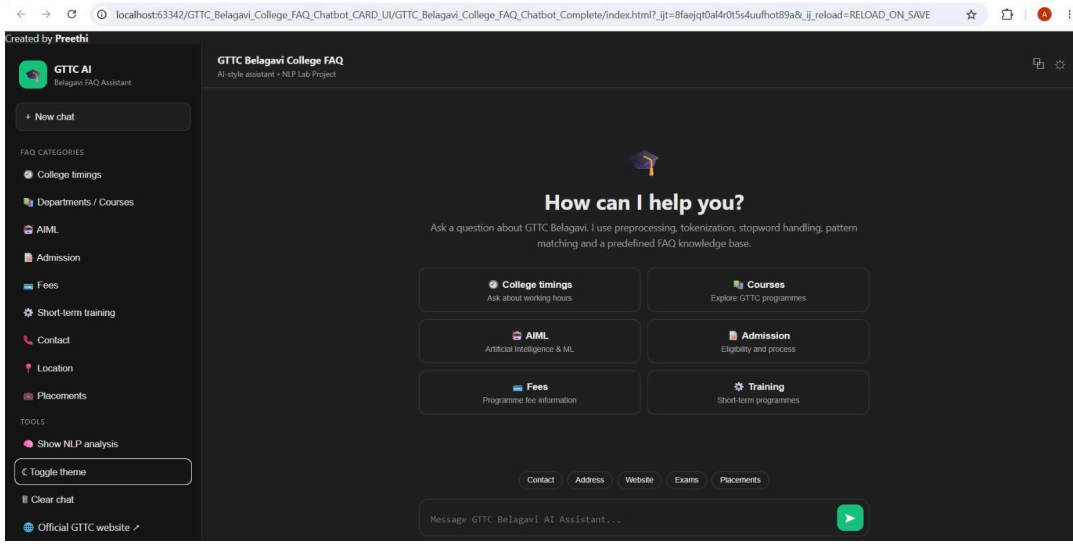
Courses

Admission

Fees

Training

Contact



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EXPLAINABLE OUTPUT

THE ANSWER COMES WITH EXPLAINABLE NLP SIGNALS

Tokens

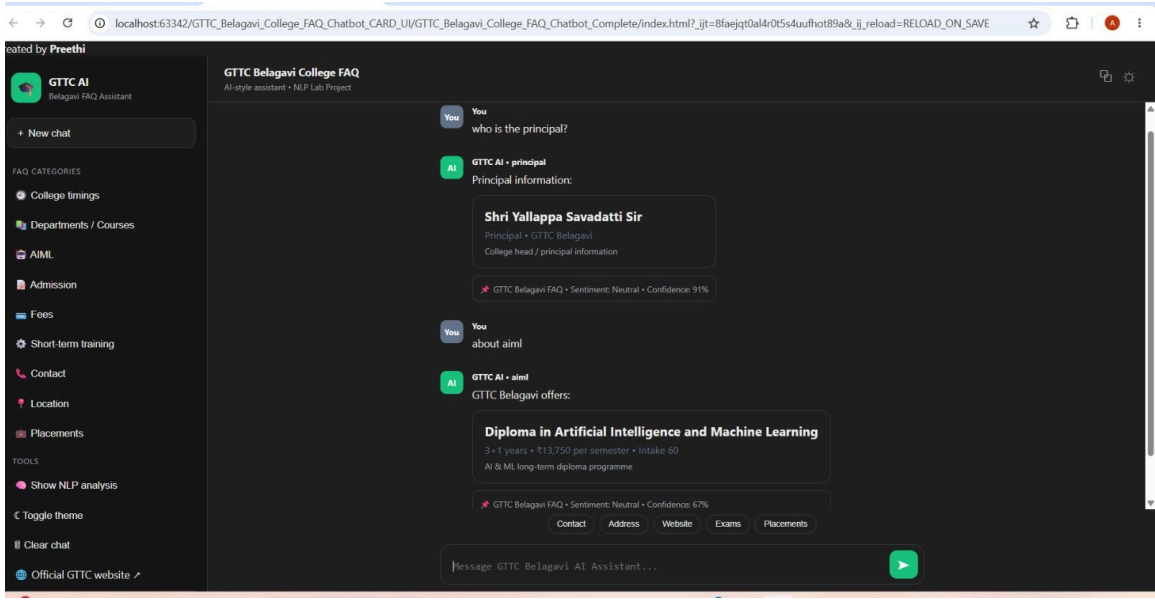
Tokens and filtered tokens show how the input was processed.

Intent

Intent and confidence make the match visible.

Sentiment

Positive, negative, or neutral in the basic analysis layer.



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IMPLEMENTATION

THE PROJECT IS SPLIT INTO A LIGHTWEIGHT WEB STACK



Python

Main NLP chatbot program



HTML and CSS

Page structure and card-style interface



JavaScript

Browser interaction and chatbot logic



CSV + regex

Sample FAQ dataset; regular expressions support tokenization and pattern matching

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TESTING

TESTING SHOWS THE FULL QUESTION-TO-ANSWER LOOP

01

Question

“What courses are available?”

02

Match

The system identifies the courses intent and searches the FAQ knowledge base.

03

Response

A predefined answer appears, with processing details in the analysis view.

Complete loop: question → intent → FAQ answer → NLP analysis

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TAKEAWAY

CONCLUSION: A PRACTICAL FOUNDATION FOR A SMARTER COLLEGE ASSISTANT

Useful foundation

Python, web technologies, and basic NLP combine into a student-facing application.

What it delivers

Quick answers, intent matching, fallback responses, and basic sentiment analysis.

Next direction

Expand the FAQ database, reduce manual updates, and move beyond rule-based matching.

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THANK YOU

Questions and discussion

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